

UK Person-Level Earnings & Cost-of-Living Data: Sources, APIs, and Access Routes

TL;DR

- **Genuinely person-level UK income/earnings microdata is NOT available through any open public API.** It sits behind the UK Data Service (registration + End User Licence or Special Licence) or, for the richest record-level files, the ONS Secure Research Service (SRS) / Integrated Data Service (IDS), which require accredited-researcher status, an institutional/organisational affiliation, and a public-good project — realistically out of reach for an unaffiliated volunteer building a public visualisation.
- **For an API-first Python project, build on aggregate official APIs** — the ONS API (no key, JSON, free) for inflation/CPIH/rent indices, the Nomis API (optional free key) for ASHE earnings by area/occupation/industry and workplace, DWP Stat-Xplore (free key) for benefits/income, and HM Land Registry Price Paid SPARQL (no key) for property prices — plus job-advert APIs (Adzuna, Reed) which are the closest thing to API-accessible, person-adjacent salary+location records.
- **The only openly downloadable person-level records are census microdata teaching files** (the 2021 Census 1% public microdata teaching sample contains 604,351 person records under the Open Government Licence) [UK Data Service](#) and synthetic populations (SPENSER); these are person-level but limited in earnings/cost-of-living detail and are files, not APIs.

Key Findings

- UK official statistics deliberately split into (a) aggregate outputs that are open/API-accessible and (b) record-level microdata that is access-controlled under the Five Safes framework and the Digital Economy Act 2017. Earnings and income are treated as disclosive personal data, so the person-level versions are gated.
- The ONS operates two relevant APIs: the older "Customise My Data"/Beta API (`api.beta.ons.gov.uk/v1`) and the newer `api.ons.gov.uk/v1` endpoint (both referenced in current ONS metadata). Both are free, need no API key, return JSON/CSV, and expose aggregate time series such as CPIH. Neither exposes person-level records.
- Nomis exposes ASHE (Annual Survey of Hours and Earnings) as aggregate estimates only — medians, means, percentiles by area, occupation, industry, and on both a workplace and residence basis — via a documented REST API. Person-level ASHE is a separate licensed product (UK Data Service SN 6689, "Annual Survey of

Hours and Earnings, 1997-2025: Secure Access", 27th Release December 2025) and is not on Nomis. The ASHE achieved sample in 2025 was 174,000 (a 1% sample of the PAYE system), per ONS "Employee earnings in the UK: 2025."

- DWP Stat-Xplore has a genuine Open Data REST API (free key) but returns aggregated tabulations of the benefits system, not record-level data.
- The best person-adjacent, truly API-first salary+location data comes from job-advert aggregators: Adzuna (free key, salary/histogram/regional endpoints) and Reed (free key, per-advert salary + location). These are job adverts, not individuals, and carry commercial terms of use.

Details

(a) Genuinely person-level microdata (access-controlled; no open API)

Labour Force Survey (LFS) / Annual Population Survey (APS) — ONS, via UK Data Service. Person-level; includes earnings, occupation, industry, hours, region. Standard End User Licence version (e.g. Quarterly LFS, SN 9433, April–June 2025 available) requires UKDS registration + EUL acceptance. A Secure Access version with finer geography/detail requires accredited-researcher status via SecureLab. No API — data downloaded as SPSS/Stata/tab files after login.

Understanding Society (UKHLS) — longitudinal person + household data with detailed income and household costs. Three tiers: End User Licence (SN 6614, most variables, download after registration); Special Licence (SN 6931, adds non-top-coded incomes, detailed SOC/SIC codes, month of birth); Secure Access (SN 6676, adds exact DOB and postcode grid references). [Understanding Society](#) Access via UK Data Service; none is Open Access; [Understanding Society](#) no API. Special Licence requires a justified application; [Understanding Society](#) Secure Access requires accredited-researcher status. The EUL version is sufficient for most researchers [Understanding Society](#) and is the only tier a solo user can realistically obtain quickly.

Living Costs and Food Survey (LCF) — the primary UK cost-of-living microdata (household expenditure + income; a two-week expenditure diary kept by all household members including children). [ons](#) Person/household-level. Available via UKDS End User Licence (latest study: 2022–2023, SN 9335). Special Licence and SRS/IDS versions exist for finer detail. [ons](#) No API. Now part of the Household Finances Survey. [ons](#)

Family Resources Survey (FRS) and derived **Households Below Average Income (HBAI)** — DWP; person/household-level income and disposable income (annual target sample ~24,000 households). [UK Data Service](#) Via UKDS (EUL/Special Licence). No API.

Wealth and Assets Survey (WAS) — ONS; household/person-level wealth, income, housing costs. Via UKDS (EUL/Secure). No API.

ASHE record-level — 1% PAYE sample (174,000 records in 2025); UKDS SN 6689 ("Secure Access"); SRS only.

ONS Longitudinal Study — ~1% of the England & Wales population linked across censuses; secure-access only via ONS, with a supplementary application; not downloadable.

Access model for the deepest tiers: the **ONS Secure Research Service (SRS)** and the **Integrated Data Service (IDS)** are Trusted Research Environments under the Digital Economy Act 2017. Both require the user to be an **accredited/approved researcher** (training + a public-good project approved by the Research Accreditation Panel), working from an approved safe setting, usually via an organisation with "Assured Organisational Connectivity." Per ONS management information last updated March 2026, there are 6,295 current Accredited Researchers and 1,575 "Current Live Accredited Researchers" actively involved in live projects, across 824 current SRS projects. Accreditation is now handled via the **People and Projects Service (PPS)** ("You can apply for researcher accreditation in the People and Projects Service (PPS)"), which — together with the Project Accreditation Service for SRS (PASS) — replaced the Research Accreditation Service closed on 4 April 2025. [UK Data Service](#) An unaffiliated individual generally cannot self-accredit without a sponsoring organisation and an approved project.

(b) API-accessible aggregate sources (recommended foundation for a Python project)

ONS API — Base URL `https://api.beta.ons.gov.uk/v1` (also `https://api.ons.gov.uk/v1`). Free, no key, no registration; JSON responses; documented rate limits (see ONS Digital blog FAQ). [Ons](#) Endpoints follow `/datasets/{id}/editions/{edition}/versions/{n}` and `/observations` with dimension filters; the `/datasets` catalogue endpoint supports `limit / offset` pagination (default 20, max 1000). Key cost-of-living datasets exposed: `cpih01` (CPIH inflation, 2005 → latest month), consumer price inflation, and `index-private-housing-rental-prices` (private rents). Downloadable as CSV with CSVW metadata (e.g. the CPIH v55 CSV at `download.ons.gov.uk/downloads/datasets/cpih01/...`). Docs: `https://developer.ons.gov.uk`. Note: some individual CMD datasets carry archive notices (e.g. the CPIH CMD page stopped updating after January 2026 data, directing users to published time series) [Office for National Statistics](#) — verify currency per dataset via `next_release`. Earnings are represented on the ONS site mainly as ASHE reference tables (better accessed via Nomis).

Example (Python):

```
import requests
url = "https://api.beta.ons.gov.uk/v1/datasets/cpih01/editions/time-series/versions/6"
r = requests.get(url).json()
```

Nomis API — Base URL

<https://www.nomisweb.co.uk/api/v01/dataset/{id}.data.csv?...> (also [.data.json](#), [.jsonstat.json](#), SDMX XML/JSON, [.data.xls](#), RSS). Optional free API key (register at <https://www.nomisweb.co.uk/myaccount/userjoin.asp>; passed as the `uid=` URL parameter). Rate/size limits: ~25,000 rows per call as a guest, ~100,000 rows per call with a key [R-universe](#) (figures from official client-library docs; some docs cite a higher absolute ceiling, so build in pagination). ASHE dataset IDs: **NM_99_1** (ASHE Workplace Analysis, mnemonic `ASHE`, data from 1998 — estimates for people working in an area) and **NM_30_1** (ASHE Resident Analysis, mnemonic `ASHER`, data from 2002 — people living in an area); latest reference year 2025 (provisional). Aggregate variables: number of jobs, median, mean, and percentiles (10, 20, 25, 30, 40, 60, 70, 75, 80, 90) across weekly/hourly/annual pay, by sex and FT/PT, each with a coefficient of variation. Geography: local authority, parliamentary constituency, region, country — and the workplace-vs-resident split gives place-of-work vs place-of-residence — but nothing below LA level. Nomis also serves LFS/APS aggregates, Census, claimant count and DWP benefits aggregates. [R-universe](#) Docs: <https://www.nomisweb.co.uk/api/v01/help>.

Example (Python):

```
import pandas as pd
url = ("https://www.nomisweb.co.uk/api/v01/dataset/NM_99_1.data.csv"
      "?
      geography=TYPE480&date=latest&sex=8&item=2&pay=1&measures=20100,20701")
df = pd.read_csv(url)
```

R users have the `nomisr` package; add a `uid` key to raise the row cap.

DWP Stat-Xplore Open Data API — Base URL [https://stat-](https://stat-xplore.dwp.gov.uk/webapi/rest/v1)

[xplore.dwp.gov.uk/webapi/rest/v1](https://stat-xplore.dwp.gov.uk/webapi/rest/v1). JSON REST; free but requires an API key [GitHub](#) (register on Stat-Xplore, key on the Account page under "Open Data API Access"); rate-limited (query the `/rate_limit` endpoint for allowance/remaining). [GitHub](#) Endpoints: `/schema` (datasets, fields, measures), `/table` (submit query, receive results), `/info`, `/rate_limit`. [dwp](#) Aggregate benefit/pensioner-income tabulations (e.g. Universal Credit monthly, Pensioners' Incomes at family/benefit-unit level). Python: `statxplore` (PyPI); R: `statxplore` (House of Commons Library), `dwpstat`. Not record-level. Docs: <https://stat-xplore.dwp.gov.uk/webapi/online-help/Open-Data-API.html>.

HM Land Registry Price Paid Data — SPARQL endpoint

<http://landregistry.data.gov.uk/landregistry/query> (also a structured URL API at [/data/ppi/...](#)). No key, no auth. [Paul-walsh](#) Every residential sale in England & Wales since January 1995 (transaction-level, not person-level): price, date, full address/postcode, property type, new-build flag, tenure, transaction category. Monthly updates (~96,000 sales/month). Open Government Licence (note Royal Mail/Ordnance Survey Address Data third-party rights restrict some commercial reuse). [service](#) Bulk

CSV also available. Python: query via `requests` + SPARQL or `SPARQLWrapper`; R: `uk1r` package. Docs: <http://landregistry.data.gov.uk/app/doc/ppd/>.

ONS private rental / house price indices — via the ONS API (`index-private-housing-rental-prices`) and the Price Index of Private Rents (PIPR, chain-linked historical series from January 2005); [ons](#) monthly.

Ofgem energy price cap — published as guidance/data tables (unit rates and standing charges by region, quarterly). No documented public REST API; download CSVs from ofgem.gov.uk. Ofgem confirms the cap rose 13% to £1,663/yr for a typical dual-fuel Direct Debit household from 1 July 2026 under new Typical Domestic Consumption Values (electricity 26.11p/kWh, standing charge 57.19p/day), per Ofgem's "Changes to energy price cap between 1 July and 30 September 2026."

Numbeo Cost of Living API — Base <https://www.numbeo.com/api/> (e.g. `/city_prices`, `/indices`, `/country_indices`, `/close_cities_with_prices`). Requires an API key (`api_key=` parameter); JSON; crowd-sourced consumer prices/rent for cities incl. London (`city_id` 6512) and other UK cities. [Numbeo](#) Paid/commercial plans and a commercial Data License; [Numbeo](#) check licensing before publishing. Python example repo: [numbeo/numbeo-api-python](#).

(c) Job-advert APIs (person-adjacent salary + location; API-first)

Adzuna API — Base <https://api.adzuna.com/v1/api/jobs/gb/...>; free App ID + key (register at developer.adzuna.com); JSON. Nine endpoints [Adzuna](#) including `/search` (per-advert: title, employer, location array, salary_min/max), [Adzuna](#) `/histogram` (salary distribution), [Adzuna](#) `/history` (historical salary trends), regional data, `/top_companies`, `/categories`, and `/jobsworth` (salary prediction from title/skills). [Adzuna](#) Genuinely UK-first and the closest API to "person-level" salary+workplace-location data (at the job-advert level). Free within rate limits (quotas on the developer portal); [Public APIs](#) commercial terms apply. Note many salaries are predicted (`salary_is_predicted` flag). A separate paid Adzuna Intelligence API (api.intelligence.adzuna.com) offers richer salary-distribution reports by location/occupation/skill/industry/employer. [Adzuna](#)

Example:

```
GET https://api.adzuna.com/v1/api/jobs/gb/search/1?
app_id=YOUR_ID&app_key=YOUR_KEY&what=python%20developer&where=london
```

Reed API — Base <https://www.reed.co.uk/api/1.0/search?...>; free API key (register on the Reed developer portal, [PublicAPI](#) used via HTTP Basic auth / `apiKey` parameter); JSON; per-advert `jobId`, `employerName`, `jobTitle`, `locationName`, `minimumSalary`/`maximumSalary`, yearly min/max, currency GBP, contract type, date. [GitHub](#) Python wrappers exist (`theo-r/reed`, `Ara225/reed-jobseeker-api`). Terms restrict redistribution; Reed's robots.txt disallows crawling `/api/`, [Apify](#) so use the

official key rather than scraping. Docs:

<https://www.reed.co.uk/developers/Jobseeker>.

Indeed / Glassdoor / LinkedIn — largely closed or partner-only APIs; not practical for an open project; scraping breaches their terms. Not recommended.

(d) Openly downloadable person-level teaching/synthetic files (person-level, but files not APIs)

2021 Census Public Microdata Teaching Sample (England & Wales) — per the ONS user guide, "a random sample of 1% of person records from Census 2021 for England and Wales; it includes records for 604,351 persons." [ons](#) 19 variables, lowest geography = region/Wales, released 7 September 2023, [ons](#) UK Data Service Study Number 9202, under the Open Government Licence and downloadable from ONS and UKDS with no registration. Person-level and openly redistributable, but limited detail and no earnings amount (income is not a census variable). Equivalents exist for Northern Ireland (NISRA, published January 2025) and Scotland (NRS), plus older teaching files (2011, 1991–1961). Safeguarded 5% individual and 1% household microdata samples [Office for National Statistics](#) require UKDS registration and EUL acceptance.

SPENSER synthetic population (Leeds Institute for Data Analytics + Alan Turing Institute, Turing-funded) [The Alan Turing Institute](#) — open-source individual-level synthetic population for Great Britain, disaggregating the 2011 Census to small areas (down to MSOA, updated toward 2020) [University of St Andrews](#) and projecting forward with dynamic microsimulation. [The Alan Turing Institute](#) Person-level records with demographic/socioeconomic attributes; useful as a synthetic microdata backbone but not a live earnings/cost-of-living source. Tools on GitHub ([alan-turing-institute/vivarium_population_spenser](#)). [GitHub](#) A published synthetic population dataset (Wu, Heppenstall, Meier, Purshouse, Lomax, 2022, *Scientific Data*, DOI 10.1038/s41597-022-01124-9) is available. [nih](#)

Comparison Table

Dataset	Provider	Person-level or aggregate	Wages	Workplace location
LFS / APS	ONS via UKDS	Person-level	Yes (earnings, occ, ind)	Region (finer in Secure)
Understanding Society (UKHLS)	ISER/Essex via UKDS	Person-level (longitudinal)	Yes	Region (LSOA in Secure)

Dataset	Provider	Person-level or aggregate	Wages	Workplace location
Living Costs & Food Survey	ONS via UKDS	Person/household	Yes (income)	Region
Family Resources Survey / HBAI	DWP via UKDS	Person/household	Yes (income)	Region
Wealth & Assets Survey	ONS via UKDS	Household/person	Yes	Region
ASHE record-level	ONS via UKDS	Person-level (1% PAYE, 174k)	Yes	Yes (workplace)
ONS Longitudinal Study	ONS	Person-level (1% linked)	Limited	Yes
ONS API	ONS	Aggregate	Limited (ASHE tables)	No
Nomis API	ONS/Durham Univ.	Aggregate	Yes (ASHE by area/occ/ind)	Yes (NM_99_1 workplace)
DWP Stat-Xplore API	DWP	Aggregate	Income/benefits	LA/region
Land Registry Price Paid	HM Land Registry	Transaction-level	No	Property location
Adzuna API	Adzuna	Job-advert level	Yes (advert salary)	Yes (advert location)
Reed API	Reed.co.uk	Job-advert level	Yes (advert salary)	Yes (advert location)
Numbeo API	Numbeo	Aggregate (city)	Salary index	City

Dataset	Provider	Person-level or aggregate	Wages	Workplace location
2021 Census Teaching File	ONS/NISRA/NRS	Person-level (604,351)	No (no income var)	Region
SPENSER synthetic pop.	Leeds/Turing	Person-level (synthetic)	Modelled	Small-area (MSOA)

Recommendations

Stage 1 — Build the visualisation on open aggregate APIs (no barriers). Combine:

- Nomis ASHE (NM_99_1 workplace, NM_30_1 resident) for earnings by area/occupation/industry —

```
pandas.read_csv("https://www.nomisweb.co.uk/api/v01/dataset/NM_99_1.data.csv?uid")
```

Register a free `uid` key to lift the row cap to ~100,000/call.
- ONS API (`cpih01` , `index-private-housing-rental-prices`) for inflation, owner-occupier housing costs, and private rents.
- HM Land Registry SPARQL for house prices by postcode/area.
- DWP Stat-Xplore (free key) for benefit/pensioner-income context.
- Ofgem tables for energy costs.

This gives a fully reproducible, redistributable (OGL) pipeline covering wages, workplace location, and cost of living without any accreditation. This should be your default architecture.

Stage 2 — Add job-advert granularity. Layer Adzuna (`/search` , `/histogram` , `/history`) and optionally Reed for near-real-time salary-by-location signals. Respect commercial terms: cache/aggregate rather than republishing raw adverts; attribute the source; check per-day/per-month rate limits before scaling. Treat Adzuna's predicted salaries with caution (see caveats).

Stage 3 — Add openly redistributable person-level texture. Use the 2021 Census public microdata teaching file (604,351 records, OGL) for genuine individual-level analysis and demographic cross-tabs, and, if you need modelled income at the individual level, SPENSER synthetic populations. These are the only person-level records you can legally publish/redistribute in a public presentation.

Stage 4 — Only if you need real record-level income microdata. Register with the UK Data Service (free) and take the EUL route for LFS / Understanding Society / LCF / FRS — but note you cannot expose raw records in a public visualisation; you may only publish derived, non-disclosive outputs. Pursue ONS SRS/IDS accreditation via the People and Projects Service only if you can secure a sponsoring organisation and an approved public-good project; realistically not feasible for a solo volunteer.

Benchmarks that change the recommendation: if you obtain a university/ONS-approved organisational affiliation → SRS/IDS record-level access becomes viable; if the project becomes commercial → re-check Adzuna/Reed/Numbeo licensing and UKDS commercial fees; if you need sub-local-authority earnings → only Secure/SRS ASHE or LFS can provide it (Nomis stops at LA level).

Caveats

- **No open API returns person-level UK earnings.** Any claim otherwise is wrong; the person-level files are gated behind UKDS licences or accreditation.
- **Licensing for redistribution:** OGL v3.0 (ONS, Nomis, Land Registry, census teaching files) permits reuse with attribution; UKDS EUL data must NOT be republished as records (only derived outputs); Adzuna/Reed/Numbeo have commercial terms restricting redistribution; Land Registry address data has Royal Mail/Ordnance Survey third-party caveats.
- **Data currency varies:** ASHE latest 2025 is provisional — ONS bulletins state "Release date: 23 October 2025" and "All estimates for 2025 are provisional and relate to the pay period that includes 30 April 2025" (median weekly full-time earnings £766.60, +5.3% nominal); LFS to April–June 2025; LCF to 2022–23; some ONS CMD API datasets carry archive notices and stop updating — always check `next_release` /version.
- **Adzuna salaries are often predicted** (flagged with `salary_is_predicted`), and job-advert data reflects vacancies, not the employed population — it is biased toward hiring sectors and should not be read as representative earnings.
- **Rate-limit figures for Nomis** (25,000 guest / 100,000 keyed rows per call) come from client-library documentation; there is minor conflict in the docs about the absolute maximum total rows, so build in pagination and error handling.
- **Numbeo and other crowd-sourced cost-of-living data** are non-official and vary in sample size per city; treat as indicative only.
- **Census teaching files contain no earnings/income variable** — the UK census does not collect income — so they are excellent for demographic/occupation cross-tabs but must be combined with ASHE (wages) and LCF/FRS (income/expenditure) for the full earnings-plus-cost-of-living picture your project needs.